Energy and Power

$$E[(x)] = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

$$P_{av} = \lim_{T\to\infty} \frac{1}{T}\int_{-T/2}^{T/2} |x(t)|^2 dt$$

- Scaling: $cy(t) = cx(t)$, sub $x(t)$ by $cx(t)$
- Additivity: sub x(t) by $x_1(t) + x_2(t)$
- Time-invariant if delaying input x(t) by T generates the same result as $y(t-T)$

BIBO

Casual System: Present value of output y(t)

Can only depend on present and past values

Ex: u(t), $2x(t-1) + x(t)$

Not Casual: $3x(t+2)$ it must know input at -2

BIBO stable if

$$\int_{-\infty}^{\infty} |h(t)| dt = finite < \infty$$

$$\cos^2(\theta) = \frac{1+\cos(\theta)}{2}$$

$$\sin^2(\theta) = \frac{1}{2}(1-\cos 2\theta)$$

$$\sin(x \pm y) = \sin(x)\cos(y) \pm \cos(x)\sin(y)$$

LTI: A system is linear if it has <u>scaling</u> and <u>additivity</u> <u>properties</u>

Differential Eqs:

$$\frac{dy}{dx} + P(x)y = Q(x)$$

$e^{\int P(x)}$ and multiply all by it

$$\sin(\theta) = \cos\left(\theta - \frac{\pi}{2}\right)$$

$$\cos(\theta) = \sin\left(\theta + \frac{\pi}{2}\right)$$

Euler's

$$\cos(\omega) = \frac{1}{2}\left(e^{j\omega} + e^{j\omega}\right)$$

$$\sin(\omega) = \frac{1}{j2}\left(e^{j\omega} - e^{j\omega}\right)$$

Odd/Even Symmetry

$$x(t) = x_e(t) + x_o(t)$$

$$x_e(t) = \frac{x(t)+x(-t)}{2},\ x_o(t) = \frac{x(t)-x(-t)}{2}$$

LCCDE: $\frac{d^2y(t)}{dt^2} + a_1\frac{dy(t)}{dt} + a_2 y(t) = x(t) + \frac{dx(t)}{dt}$

$$\alpha = \frac{a_1}{2}\ ;\ \omega_o = \sqrt{a_2}\ ;\ \varsigma = \frac{\alpha}{\omega_o}$$

Characteristic eq: $s^2 + a_1 s + a_2 = 0$

$a_1^2 > 4a_2$ *Both roots real(overdamped)*

$a_1^2 < 4a_2$ *complex conjugate roots (underdamped)*

$a_1^2 = 4a_2$ *identical roots (critically damped)*

Frequency Response: $\widehat{H}(\omega)$

$$y(t) = \widehat{H}(\omega)e^{j\omega t}\ ;\ x(t) = e^{j\omega t}$$

$$\frac{d}{dt} \to j\omega\ ;\ \frac{d^2}{dt^2} \to (j\omega)^2$$

$$h(t) = e^{-t}u(t)\ ;\ \widehat{H}(\omega) = \int_{-\infty}^{\infty} h(t)e^{-j\omega t}dt = \frac{1}{1+j\omega}$$

Laplace Transform:

$$X(s) = L[x(t)] = \int_{0^-}^{\infty} x(t)e^{-st}dt$$

$$s = \sigma + j\omega\ ;\ |s| = \sqrt{\sigma^2 + \omega^2}$$

Initial-Value Th: $x(0^+) = \lim_{s\to\infty} sX(s)$

Final-Value Th: $x(\infty) = \lim_{s\to 0} sX(s)$

Laplace Transform Pairs

	$x(t)$		$\mathbf{X(s)} = \mathcal{L}[x(t)]$
1	$\delta(t)$	⟷	1
1a	$\delta(t-T)$	⟷	e^{-Ts}
2	$u(t)$	⟷	$\frac{1}{\mathbf{s}}$
2a	$u(t-T)$	⟷	$\frac{e^{-Ts}}{\mathbf{s}}$
3	$e^{-at}\,u(t)$	⟷	$\frac{1}{\mathbf{s}+a}$
3a	$e^{-a(t-T)}\,u(t-T)$	⟷	$\frac{e^{-Ts}}{\mathbf{s}+a}$
4	$t\,u(t)$	⟷	$\frac{1}{\mathbf{s}^2}$
4a	$(t-T)\,u(t-T)$	⟷	$\frac{e^{-Ts}}{\mathbf{s}^2}$
5	$t^2\,u(t)$	⟷	$\frac{2}{\mathbf{s}^3}$
6	$te^{-at}\,u(t)$	⟷	$\frac{1}{(\mathbf{s}+a)^2}$
7	$t^2e^{-at}\,u(t)$	⟷	$\frac{2}{(\mathbf{s}+a)^3}$
8	$t^{n-1}e^{-at}\,u(t)$	⟷	$\frac{(n-1)!}{(\mathbf{s}+a)^n}$
9	$\sin(\omega_0 t)\,u(t)$	⟷	$\frac{\omega_0}{\mathbf{s}^2+\omega_0^2}$
10	$\sin(\omega_0 t+\theta)\,u(t)$	⟷	$\frac{\mathbf{s}\sin\theta+\omega_0\cos\theta}{\mathbf{s}^2+\omega_0^2}$
11	$\cos(\omega_0 t)\,u(t)$	⟷	$\frac{\mathbf{s}}{\mathbf{s}^2+\omega_0^2}$
12	$\cos(\omega_0 t+\theta)\,u(t)$	⟷	$\frac{\mathbf{s}\cos\theta-\omega_0\sin\theta}{\mathbf{s}^2+\omega_0^2}$
13	$e^{-at}\sin(\omega_0 t)\,u(t)$	⟷	$\frac{\omega_0}{(\mathbf{s}+a)^2+\omega_0^2}$
14	$e^{-at}\cos(\omega_0 t)\,u(t)$	⟷	$\frac{\mathbf{s}+a}{(\mathbf{s}+a)^2+\omega_0^2}$
15	$2e^{-at}\cos(bt-\theta)\,u(t)$	⟷	$\frac{e^{j\theta}}{\mathbf{s}+a+jb}+\frac{e^{-j\theta}}{\mathbf{s}+a-jb}$
15a	$e^{-at}\cos(bt-\theta)\,u(t)$	⟷	$\frac{(\mathbf{s}+a)\cos\theta+b\sin\theta}{(\mathbf{s}+a)^2+b^2}$
16	$\frac{2t^{n-1}}{(n-1)!}e^{-at}\cos(bt-\theta)\,u(t)$	⟷	$\frac{e^{j\theta}}{(\mathbf{s}+a+jb)^n}+\frac{e^{-j\theta}}{(\mathbf{s}+a-jb)^n}$

Property	$x(t)$		$\mathbf{X(s)} = \mathcal{L}[x(t)]$
1. Multiplication by constant	$K\,x(t)$	⟷	$K\,\mathbf{X(s)}$
2. Linearity	$K_1\,x_1(t)+K_2\,x_2(t)$	⟷	$K_1\,\mathbf{X}_1(\mathbf{s})+K_2\,\mathbf{X}_2(\mathbf{s})$
3. Time scaling	$x(at),\quad a>0$	⟷	$\frac{1}{a}\mathbf{X}\left(\frac{\mathbf{s}}{a}\right)$
4. Time shift	$x(t-T)\,u(t-T)$	⟷	$e^{-T\mathbf{s}}\,\mathbf{X(s)},\quad T\geq 0$
5. Frequency shift	$e^{-\mathbf{a}t}\,x(t)$	⟷	$\mathbf{X}(\mathbf{s}+\mathbf{a})$
6. Time 1st derivative	$x' = \frac{dx}{dt}$	⟷	$\mathbf{s}\,\mathbf{X(s)} - x(0^-)$
7. Time 2nd derivative	$x'' = \frac{d^2x}{dt^2}$	⟷	$\mathbf{s}^2\mathbf{X(s)} - \mathbf{s}x(0^-) - x'(0^-)$
8. Time integral	$\int_{0^-}^{t} x(t')\,dt'$	⟷	$\frac{1}{\mathbf{s}}\mathbf{X(s)}$
9. Frequency derivative	$t\,x(t)$	⟷	$-\frac{d}{d\mathbf{s}}\mathbf{X(s)} = -\mathbf{X}'(\mathbf{s})$
10. Frequency integral	$\frac{x(t)}{t}$	⟷	$\int_{\mathbf{s}}^{\infty}\mathbf{X}(\mathbf{s}')\,d\mathbf{s}'$
11. Initial value	$x(0^+)$	=	$\lim_{\mathbf{s}\to\infty}\mathbf{s}\,\mathbf{X(s)}$
12. Final value	$\lim_{t\to\infty} x(t) = x(\infty)$	=	$\lim_{\mathbf{s}\to 0}\mathbf{s}\,\mathbf{X(s)}$
13. Convolution	$x_1(t) * x_2(t)$	⟷	$\mathbf{X}_1(\mathbf{s})\,\mathbf{X}_2(\mathbf{s})$

$m = n, X(s)$ proper

$m > n, X(s)$ improper

$m < n, X(s)$ strictly proper

$$\frac{N(s)}{D(s)} = 1 + \frac{N(s)-D(s)}{D(s)}$$

Complete Square:

$$x^2 + bx + c = x^2 + bx + \left(\frac{b}{2}\right)^2 + c - \left(\frac{b}{2}\right)^2$$

$$= \left(x + \frac{b}{2}\right)^2 + c - \left(\frac{b}{2}\right)^2$$

Repeated Poles:

$$B_m = X(s) * (s-p)^m|_{s=p}$$

$$B_{m-1} = \frac{d}{ds}X(s) * (s-p)^m|_{s=p}$$

$$B_{m-2} = \frac{1}{2} * \frac{d^2}{ds^2}X(s) * (s-p)^m|_{s=p}$$

$$ax + by = e$$
$$cx + dy = f$$

$$\Delta = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc \neq 0$$ IMP

$$\Delta_x = \begin{vmatrix} e & b \\ f & d \end{vmatrix} = ed - bf,\quad \Delta_y = \begin{vmatrix} a & e \\ c & f \end{vmatrix} = af - ec$$

$$x = \frac{\Delta_x}{\Delta},\quad y = \frac{\Delta_y}{\Delta}$$

Miscellaneous

$$tu(t-1) \to (t-1)u(t-1) + u(t-1)$$

$$\delta(at) = \frac{1}{a}\delta(t)$$

$$(t-1)e^{-t} = e^{-1}(t-1)e^{-(t-1)}$$

Forced response: terms not in form $\frac{1}{s+a}$

Modes: The roots of the Characteristic Poly.

$$s^2 + 3s + 2 = 0 \to (S+1)(s+2) = 0$$

$$s = -1, -2$$

Integration by parts

$$\int u dv = uv - \int v du$$

Check for Invertibility: $\frac{1}{H(s)}$

Proper m=n

Zeros on OLHP

Poles on OLHP

$$\cos(x \pm y) = \cos(x)\cos(y) \mp \sin(x)\sin(y)$$

Real (-) pole results in decaying e^{pt}

Real (+) pole results in growing e^{pt}

Partitions:

$V(s) = \frac{RC}{1+sRC}V(0^-) + \frac{1}{1+sRC}V_i(s)$ ZIR: set $V_i(s) = 0$

ZIR ZSR ZSR: Zero initial conditions

Transient: Decay to 0 overtime

Steady-State: Remains overtime

Natural Response: Terms that depend only on MODES

Forced Response: All remaining terms

$$\mathbf{X(s)} = \frac{\mathbf{N(s)}}{\mathbf{D(s)}} = C\frac{(\mathbf{s}-z_1)(\mathbf{s}-z_2)\ldots(\mathbf{s}-z_m)}{(\mathbf{s}-p_1)(\mathbf{s}-p_2)\ldots(\mathbf{s}-p_n)}$$

The orders of the numerator (*m*) and denominator (*n*) polynomials are important

$m < n$: Strictly proper **MOST CONVENIENT**

$m = n$: Proper

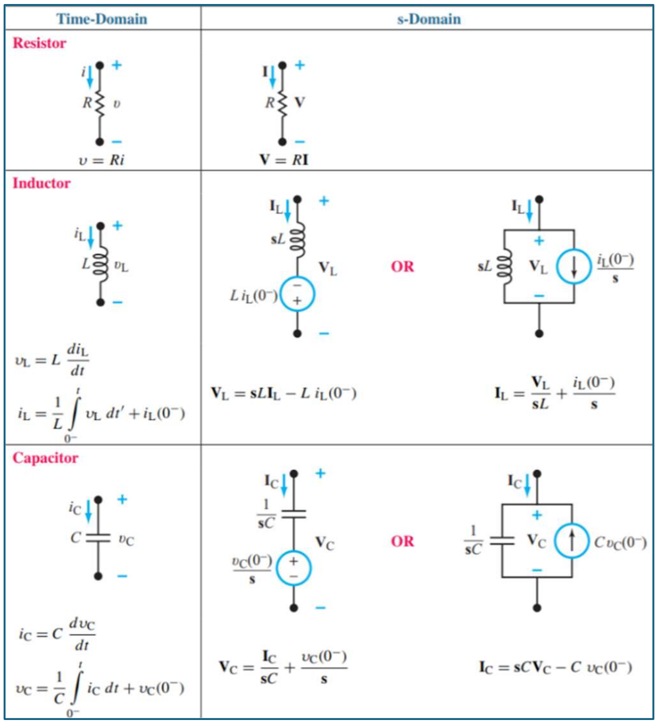
$m > n$: Improper

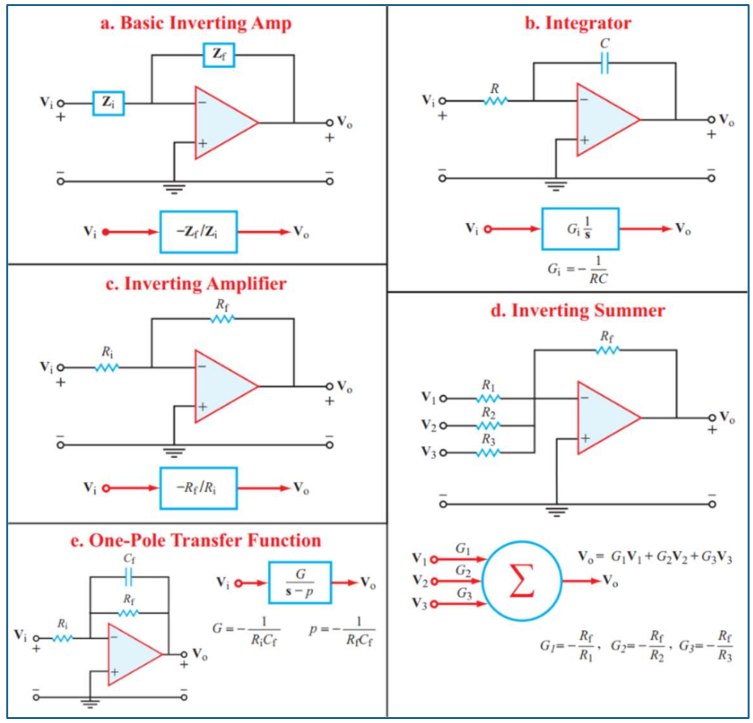
Let's work through some examples to get the hang of it

3-11.2 Natural / Forced Partition

If we partition the total response into a

- **Natural response** = all terms that vary with t according to the system's modes (in the present case, the system has only one mode, namely e^{-3t}), and a
- **Forced response** = all remaining terms,





Recap: Trigonometric Identities

$\sin(x \pm y) = \sin x\cos y \pm \cos x\sin y$ (basic formula 1)

$\cos(x \pm y) = \cos x\cos y \mp \sin x\sin y$ (basic formula 2)

$\Rightarrow \sin(2x) = 2\sin x\cos x$

$\Rightarrow \cos(2x) = \cos^2 x - \sin^2 x$

$\Rightarrow \sin x\sin y = \frac{1}{2}[\cos(x-y) - \cos(x+y)]$

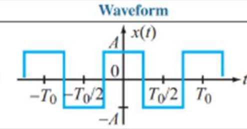
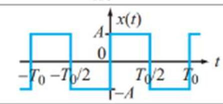
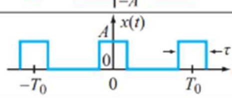
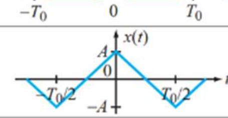
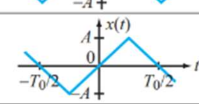
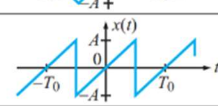
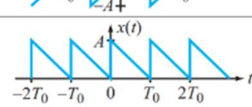
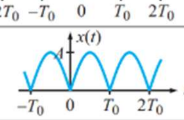
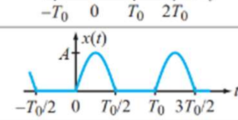
$\Rightarrow \cos x\cos y = \frac{1}{2}[\cos(x-y) + \cos(x+y)]$

ETC.

$$2\cos(a)\cos(b) = \cos(a+b) + \cos(a-b)$$

Integral
$\int_0^{T_0} \sin(n\omega_0 t + \phi)\,dt = 0$
$\int_0^{T_0} \cos(n\omega_0 t + \phi)\,dt = 0$
$\int_0^{T_0} \sin(n\omega_0 t + \phi_1)\sin(m\omega_0 t + \phi_2)\,dt = 0 \quad n \neq m$
$\int_0^{T_0} \cos(n\omega_0 t + \phi_1)\cos(m\omega_0 t + \phi_2)\,dt = 0 \quad n \neq m$
$\int_0^{T_0} \sin(n\omega_0 t + \phi_1)\cos(m\omega_0 t + \phi_2)\,dt = 0$
$\int_0^{T_0} \sin^2(n\omega_0 t + \phi)\,dt = T_0/2$
$\int_0^{T_0} \cos^2(n\omega_0 t + \phi)\,dt = T_0/2$
$\int_0^{T_0} \cos(n\omega_0 t + \phi_1)\cos(n\omega_0 t + \phi_2)\,dt = \frac{T_0}{2}\cos(\phi_1 - \phi_2)$

Figure 5.4.2: Fourier series expressions for a select set of periodic waveforms.

	Waveform	Fourier Series
1. Square Wave		$x(t) = \sum_{n=1}^{\infty} \frac{4A}{n\pi}\sin\left(\frac{n\pi}{2}\right)\cos\left(\frac{2n\pi t}{T_0}\right)$
2. Time-Shifted Square Wave		$x(t) = \sum_{\substack{n=1 \\ n=\text{odd}}}^{\infty} \frac{4A}{n\pi}\sin\left(\frac{2n\pi t}{T_0}\right)$
3. Pulse Train		$x(t) = \frac{A\tau}{T_0} + \sum_{n=1}^{\infty} \frac{2A}{n\pi}\sin\left(\frac{n\pi\tau}{T_0}\right)\cos\left(\frac{2n\pi t}{T_0}\right)$
4. Triangular Wave		$x(t) = \sum_{\substack{n=1 \\ n=\text{odd}}}^{\infty} \frac{8A}{n^2\pi^2}\cos\left(\frac{2n\pi t}{T_0}\right)$
5. Shifted Triangular Wave		$x(t) = \sum_{\substack{n=1 \\ n=\text{odd}}}^{\infty} \frac{8A}{n^2\pi^2}\sin\left(\frac{n\pi}{2}\right)\sin\left(\frac{2n\pi t}{T_0}\right)$
6. Sawtooth		$x(t) = \sum_{n=1}^{\infty} (-1)^{n+1}\frac{2A}{n\pi}\sin\left(\frac{2n\pi t}{T_0}\right)$
7. Backward Sawtooth		$x(t) = \frac{A}{2} + \sum_{n=1}^{\infty} \frac{A}{n\pi}\sin\left(\frac{2n\pi t}{T_0}\right)$
8. Full-Wave Rectified Sinusoid		$x(t) = \frac{2A}{\pi} + \sum_{n=1}^{\infty} \frac{4A}{\pi(1-4n^2)}\cos\left(\frac{2n\pi t}{T_0}\right)$
9. Half-Wave Rectified Sinusoid		$x(t) = \frac{A}{\pi} + \frac{A}{2}\sin\left(\frac{2\pi t}{T_0}\right) + \sum_{\substack{n=2 \\ n=\text{even}}}^{\infty} \frac{2A}{\pi(1-n^2)}\cos\left(\frac{2n\pi t}{T_0}\right)$

Cosine/Sine	Amplitude/Phase	Exponential
$x(t) = a_0 + \sum_{n=1}^{\infty}[a_n\cos(n\omega_0 t) + b_n\sin(n\omega_0 t)]$	$x(t) = c_0 + \sum_{n=1}^{\infty} c_n\cos(n\omega_0 t + \phi_n)$	$x(t) = \sum_{n=-\infty}^{\infty} \mathbf{x}_n e^{jn\omega_0 t}$
$a_0 = \frac{1}{T_0}\int_0^{T_0} x(t)\,dt$	$c_n e^{j\phi_n} = a_n - jb_n$	$\mathbf{x}_n = \lvert\mathbf{x}_n\rvert e^{j\phi_n};\ \mathbf{x}_{-n} = \mathbf{x}_n^*;\ \phi_{-n} = -\phi_n$
$a_n = \frac{2}{T_0}\int_0^{T_0} x(t)\cos n\omega_0 t\,dt$	$c_n = \sqrt{a_n^2 + b_n^2}$	$\lvert\mathbf{x}_n\rvert = c_n/2;\ x_0 = c_0$
$b_n = \frac{2}{T_0}\int_0^{T_0} x(t)\sin n\omega_0 t\,dt$	$\phi_n = \begin{cases} -\tan^{-1}(b_n/a_n), & a_n > 0 \\ \pi - \tan^{-1}(b_n/a_n), & a_n < 0 \end{cases}$	$\mathbf{x}_n = \frac{1}{T_0}\int_0^{T_0} x(t)\,e^{-jn\omega_0 t}\,dt$
$a_0 = c_0 = x_0;$	$a_n = c_n\cos\phi_n;\ b_n = -c_n\sin\phi_n;$	$\mathbf{x}_n = \frac{1}{2}(a_n - jb_n)$

$a_n = 0$ if waveform has odd Symmetry

$b_n = 0$ if waveform has even Symmetry

3-11.3 Transient / Steady-State Partition

In many practical situations, the system analyst or designer may be particularly interested in partitioning the response into a

- **Transient response** = all terms that decay to zero as $t \to \infty$, and a
- **Steady-state response** = all terms that remain after the demise of the transient response.

Parseval's Theorem:

$$\text{Instantaneous Power} = p(t) = |x(t)|^2 = x(t)x^*(t)$$

Total Energy E Expended from t_1 to t_2:

$$E = \int_{t_1}^{t_2} p(t)\,dt = \int_{t_1}^{t_2} |x(t)|^2\,dt = \int_{t_1}^{t_2} x(t)x^*(t)\,dt$$

$$\text{Average Power: } P_{av} = \lim_{T\to\infty}\frac{1}{T}\int_{-T/2}^{+T/2} p(t)\,dt = \lim_{T\to\infty}\frac{1}{T}\int_{-T/2}^{+T/2} |x(t)|^2\,dt$$

Average Power of a Sinusoid

$$x_1(t) = A\cos(n\omega_0 t + \phi), \quad \omega_0 \neq 0, \quad n > 0$$

$$P_{x1} = \frac{A^2}{2}$$

Power of a Complex Exponential

$$x_2(t) = Ae^{j(\omega_0 t + \phi)}$$

$$P_{x2} = A^2$$

3 Versions of Power Formula

$$P_x = \frac{1}{T_0}\int_0^{T_0} |x(t)|^2\,dt = a_0^2 + \sum_{n=1}^{\infty}(a_n^2 + b_n^2)/2, \quad (5.58a)$$

$$P_x = \frac{1}{T_0}\int_0^{T_0} |x(t)|^2\,dt = c_0^2 + \sum_{n=1}^{\infty} c_n^2/2, \quad (5.58b)$$

$$P_x = \frac{1}{T_0}\int_0^{T_0} |x(t)|^2\,dt = \sum_{n=-\infty}^{\infty} |\mathbf{x}_n|^2. \quad (5.58c)$$

$$P_{x+y} = \frac{c_n^2}{2} + \frac{c_m^2}{2}$$

- The power associated with the zero frequency terms (a_0, c_0, or $|x_0|$) is called the **dc power**
- The remaining power associated with the harmonics is referred to as the **ac power**
- The **ac power fraction** is the ratio of the *ac* power and the sum of the *ac* power and *dc* power

The Fourier Transform

$$\hat{\mathbf{X}}(\omega) = \mathcal{F}[x(t)] = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}\,dt$$

$$x(t) = \mathcal{F}^{-1}[\hat{\mathbf{X}}(\omega)] = \frac{1}{2\pi}\int_{-\infty}^{\infty} \hat{\mathbf{X}}(\omega)e^{j\omega t}\,d\omega$$

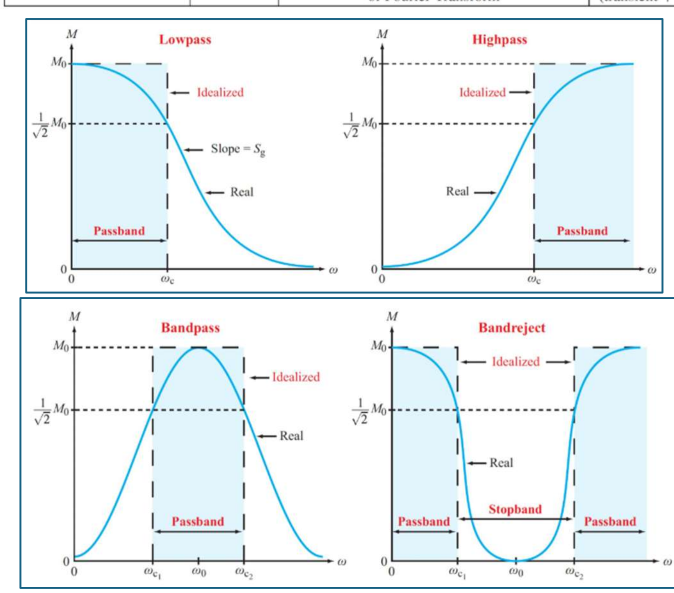
- **Zero-input response** (**ZIR**) = response of the system in the absence of an input excitation ($\mathbf{V}_i(\mathbf{s}) = 0$).
- **Zero-state response** (**ZSR**) = response of the system when it is in zero state (zero initial conditions, $A_c = \upsilon(0^-) = 0$).

	$x(t)$		$\mathbf{X}(\omega) = \mathcal{F}[x(t)]$	$\lvert\mathbf{X}(\omega)\rvert$
1.	$\delta(t)$	↔	1	
1a.	$\delta(t - t_0)$	↔	$e^{-j\omega t_0}$	
2.	1	↔	$2\pi\,\delta(\omega)$	2π
3.	$u(t)$	↔	$\pi\,\delta(\omega) + 1/j\omega$	π
4.	$\mathrm{sgn}(t)$	↔	$2/j\omega$	
5.	$\mathrm{rect}(t/\tau)$	↔	$\tau\,\mathrm{sinc}(\omega\tau/2)$	τ; $\frac{-2\pi}{\tau}$, $\frac{2\pi}{\tau}$
6.	$\frac{e^{-t^2/(2\sigma^2)}}{\sqrt{2\pi\sigma^2}}$	↔	$e^{-\omega^2\sigma^2/2}$	
7a.	$e^{-at}\,u(t)$	↔	$1/(a + j\omega)$	$1/a$
7b.	$e^{at}\,u(-t)$	↔	$1/(a - j\omega)$	$1/a$
8.	$\cos\omega_0 t$	↔	$\pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$	π, π; $-\omega_0$, ω_0
9.	$\sin\omega_0 t$	↔	$j\pi[\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$	$j\pi$, $-j\pi$; $-\omega_0$, ω_0
10.	$e^{j\omega_0 t}$	↔	$2\pi\,\delta(\omega - \omega_0)$	
11.	$te^{-at}\,u(t)$	↔	$1/(a + j\omega)^2$	
12a.	$[e^{-at}\sin\omega_0 t]\,u(t)$	↔	$\omega_0/[(a + j\omega)^2 + \omega_0^2]$	
12b.	$[\sin\omega_0 t]\,u(t)$	↔	$(\pi/2j)[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)] + [\omega_0^2/(\omega_0^2 - \omega^2)]$	
13a.	$[e^{-at}\cos\omega_0 t]\,u(t)$	↔	$(a + j\omega)/[(a + j\omega)^2 + \omega_0^2]$	
13b.	$[\cos\omega_0 t]\,u(t)$	↔	$(\pi/2)[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)] + [j\omega/(\omega_0^2 - \omega^2)]$	

Fourier transform properties.

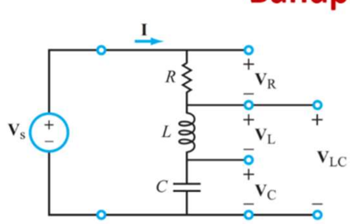
Property	$x(t)$		$\mathbf{X}(\omega) = \mathcal{F}[x(t)] = \int_{-\infty}^{\infty} x(t)\,e^{-j\omega t}\,dt$
1. Multiplication by constant	$K\,x(t)$	↔	$K\,\mathbf{X}(\omega)$
2. Linearity	$K_1\,x_1(t) + K_2\,x_2(t)$	↔	$K_1\,\mathbf{X}_1(\omega) + K_2\,\mathbf{X}_2(\omega)$
3. Time scaling	$x(at)$	↔	$\frac{1}{\lvert a\rvert}\mathbf{X}\left(\frac{\omega}{a}\right)$
4. Time shift	$x(t - t_0)$	↔	$e^{-j\omega t_0}\,\mathbf{X}(\omega)$
5. Frequency shift	$e^{j\omega_0 t}\,x(t)$	↔	$\mathbf{X}(\omega - \omega_0)$
6. Time 1st derivative	$x' = \frac{dx}{dt}$	↔	$j\omega\,\mathbf{X}(\omega)$
7. Time nth derivative	$\frac{d^n x}{dt^n}$	↔	$(j\omega)^n\,\mathbf{X}(\omega)$
8. Time integral	$\int_{-\infty}^{t} x(\tau)\,d\tau$	↔	$\frac{\mathbf{X}(\omega)}{j\omega} + \pi\,\mathbf{X}(0)\,\delta(\omega)$, where $\mathbf{X}(0) = \int_{-\infty}^{\infty} x(t)\,dt$
9. Frequency derivative	$t^n\,x(t)$	↔	$(j)^n\frac{d^n\mathbf{X}(\omega)}{d\omega^n}$
10. Modulation	$x(t)\cos\omega_0 t$	↔	$\frac{1}{2}[\mathbf{X}(\omega - \omega_0) + \mathbf{X}(\omega + \omega_0)]$
11. Convolution in t	$x_1(t) * x_2(t)$	↔	$\mathbf{X}_1(\omega)\,\mathbf{X}_2(\omega)$
12. Convolution in ω	$x_1(t)\,x_2(t)$	↔	$\frac{1}{2\pi}\mathbf{X}_1(\omega) * \mathbf{X}_2(\omega)$
13. Conjugate symmetry			$\mathbf{X}(-\omega) = \mathbf{X}^*(\omega)$

Input $x(t)$		Solution Method	Output $y(t)$
Duration	Waveform		
Everlasting	Sinusoid	Phasor Domain	Steady State Component (no transient exists)
Everlasting	Periodic	Phasor Domain and Fourier Series	Steady State Component (no transient exists)
Causal, $x(t) = 0$, for $t < 0$	Any	Laplace Transform (unilateral) (can accommodate non-zero initial conditions)	Complete Solution (transient + steady state)
Everlasting	Any	Bilateral Laplace Transform or Fourier Transform	Complete Solution (transient + steady state)



bandstop filtering, we choose V_{LC}

Bandpass Filter



$$B = \omega_{c2} - \omega_{c1} = \frac{R}{L} \quad \text{BANDWIDTH}$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

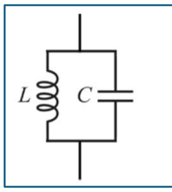
Maximum value at $\omega_0 = 1$

$$\text{Half power} \Rightarrow M_{BP}(\omega) = \frac{1}{\sqrt{2}}$$

Bandpass Filter $\hat{\mathbf{H}}_R(\omega) = \frac{\mathbf{V}_R}{\mathbf{V}_S} = \frac{\mathbf{I}R}{\mathbf{V}_S} = \frac{j\omega RC}{j\omega RC + (1-\omega^2 LC)}$ when $\omega \to 0 \Rightarrow |\hat{\mathbf{H}}_R(\omega)| \to 0$; when $\omega \to \infty \Rightarrow |\hat{\mathbf{H}}_R(\omega)| \to 0$

HIGH-PASS $\hat{\mathbf{H}}_L(\omega) = \frac{\mathbf{V}_L}{\mathbf{V}_S} = \frac{(j\omega L)\mathbf{I}}{\mathbf{V}_S} = \frac{(j\omega L)(j\omega C)}{j\omega RC + (1-\omega^2 LC)}$ when $\omega \to 0 \Rightarrow |\hat{\mathbf{H}}_L(\omega)| \to 0$; when $\omega \to \infty \Rightarrow |\hat{\mathbf{H}}_L(\omega)| \to 1$

LOW-PASS FILTER $\hat{\mathbf{H}}_C(\omega) = \frac{\mathbf{V}_C}{\mathbf{V}_S} = \frac{\mathbf{I}}{j\omega C} \times \frac{1}{\mathbf{V}_S} = \frac{1}{j\omega RC + (1-\omega^2 LC)}$ when $\omega \to 0 \Rightarrow |\hat{\mathbf{H}}_C(\omega)| \to 1$; when $\omega \to \infty \Rightarrow |\hat{\mathbf{H}}_C(\omega)| \to 0$



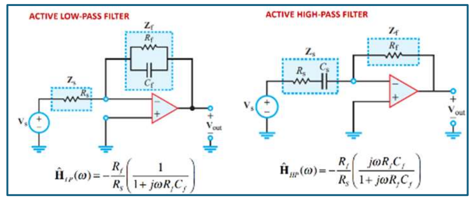
$Z = Z_L || Z_C = \frac{Z_L Z_C}{Z_L + Z_C}$

$Z \to \infty$ as $\omega \to \omega_0$ since

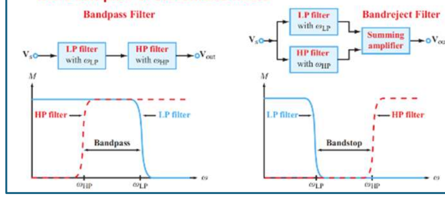
$Z_L + Z_C \to 0$

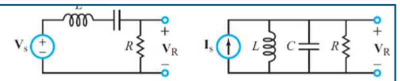
$\text{sinc}(t) \Longleftrightarrow \pi\, \text{rect}\left(\frac{\omega}{2}\right)$

$e^{j\theta} = \cos\theta + j\sin\theta$



Basic filters can be combined in series and/or parallel to build more complex or customized filters





RLC circuit		
Transfer function	$\mathbf{H} = \frac{\mathbf{V}_R}{\mathbf{V}_s}$	$\mathbf{H} = \frac{\mathbf{V}_R}{\mathbf{I}_s}$
Resonant frequency, ω_0	$\frac{1}{\sqrt{LC}}$	$\frac{1}{\sqrt{LC}}$
Bandwidth, B	$\frac{R}{L}$	$\frac{1}{RC}$
Quality factor, Q	$\frac{\omega_0}{B} = \frac{\omega_0 L}{R}$	$\frac{\omega_0}{B} = \frac{R}{\omega_0 L}$

$x(t) \xrightarrow[\text{(Replace } t \text{ by } (t-b))]{\text{Time shift by } b} x(t-b)$

$x(t-b) \xrightarrow[\text{(Replace } t \text{ by } (at))]{\text{Time scale by } a} x(at-b)$

S.No.	Form of the rational function	Form of the partial fraction
1.	$\frac{px+q}{(x-a)(x-b)}, a \neq b$	$\frac{A}{x-a} + \frac{B}{x-b}$
2.	$\frac{px+q}{(x-a)^2}$	$\frac{A}{x-a} + \frac{B}{(x-a)^2}$
3.	$\frac{px^2+qx+r}{(x-a)(x-b)(x-c)}$	$\frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{x-c}$
4.	$\frac{px^2+qx+r}{(x-a)^2(x-b)}$	$\frac{A}{x-a} + \frac{B}{(x-a)^2} + \frac{C}{x-b}$
5.	$\frac{px^2+qx+r}{(x-a)(x^2+bx+c)}$	$\frac{A}{x-a} + \frac{Bx+C}{x^2+bx+c}$
	• where $x^2 + bx + c$ cannot be factorised further	

$$M = |\hat{\mathbf{H}}(\omega)| = \frac{(\omega/\omega_C)}{\sqrt{1+(\omega/\omega_C)^2}}$$

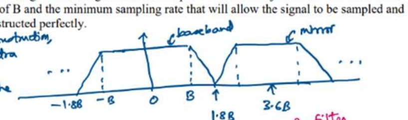
$$M\,[\text{dB}] = 20\log M = \underbrace{20\log(\omega/\omega_C)}_{1} - \underbrace{20\log\sqrt{1+(\omega/\omega_C)^2}}_{2}$$

$$\hat{\mathbf{H}}(\omega) = \frac{j(\omega/\omega_C)}{1+j(\omega/\omega_C)}$$

$$\phi(\omega) = \phi_{\text{numerator}} - \phi_{\text{denominator}} = \underbrace{90^o}_{1} - \underbrace{\tan^{-1}(\omega/\omega_C)}_{2}$$

Sketch a diagram showing how these components will be adjusted and interconnected. Find the value of B and the minimum sampling rate that will allow the signal to be sampled and reconstructed perfectly.

For perfect reconstruction, the mirror spectra should not overlap with the baseband spectrum



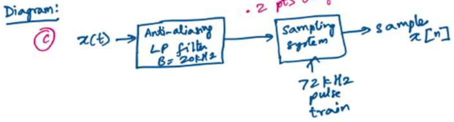
$\Rightarrow B = 20$ kHz Ⓐ setting for LP filter

sampling rate $= 3.6 \times 20 = 72$ kHz Ⓑ setting for Sampling system

Grading
- 2 pts trying
- +3 pts each Ⓐ, Ⓑ
- 2 pts diagram Ⓒ

Diagram:

Ⓒ



Summary: Application to Circuits

Step 1: Express $v_s(t)$ in terms of an amplitude/phase Fourier Series:

$$\upsilon_S(t) = a_0 + \sum_{n=1}^{\infty} c_n \cos(n\omega_0 t + \phi_n) \quad (1)$$

Step 2: Determine the transfer function of the circuit at frequency ω:

$$\hat{\mathbf{H}}(\omega) = \mathbf{V}_{out} \quad \text{when } \upsilon_S = \cos\omega t$$

Step 3: Adjust each term in equation (1) by the transfer function from Step 2:

$$\upsilon_{out}(t) = a_0\hat{\mathbf{H}}(0) + \sum_{n=1}^{\infty} c_n \left|\hat{\mathbf{H}}(n\omega_0)\right| \cos(n\omega_0 + \phi_n + \angle\hat{\mathbf{H}}(n\omega_0))$$

a. (5 points) An LTI system has the impulse response $h(t) = \cos(t)u(t)$. Calculate the response to the input signal $x(t) = \cos(t)\,u(t)$ and determine whether or not this system is BIBO stable.

Time-domain solution: $y(t) = \cos(t)\,u(t) * \cos(t)\,u(t)$ Ⓐ

$= \frac{(e^{jt}+e^{-jt})}{2} u(t) * \frac{(e^{jt}+e^{-jt})}{2} u(t)$

$= \frac{1}{4}\{e^{jt}u(t) * e^{jt}u(t) + e^{jt}u(t) * e^{-jt}u(t) + e^{-jt}u(t) * e^{jt}u(t) + e^{-jt}u(t) * e^{-jt}u(t)\}$

3 $= \frac{1}{4}\{te^{jt} + 2\left(\frac{e^{jt}-e^{-jt}}{2j}\right) + te^{-jt}\}u(t) = \underbrace{\frac{1}{2}t\cos(t)\,u(t)}_{\text{unbounded}} + \frac{1}{2}\sin(t)u(t)$

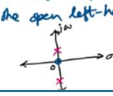
$\Rightarrow$ NOT BIBO stable

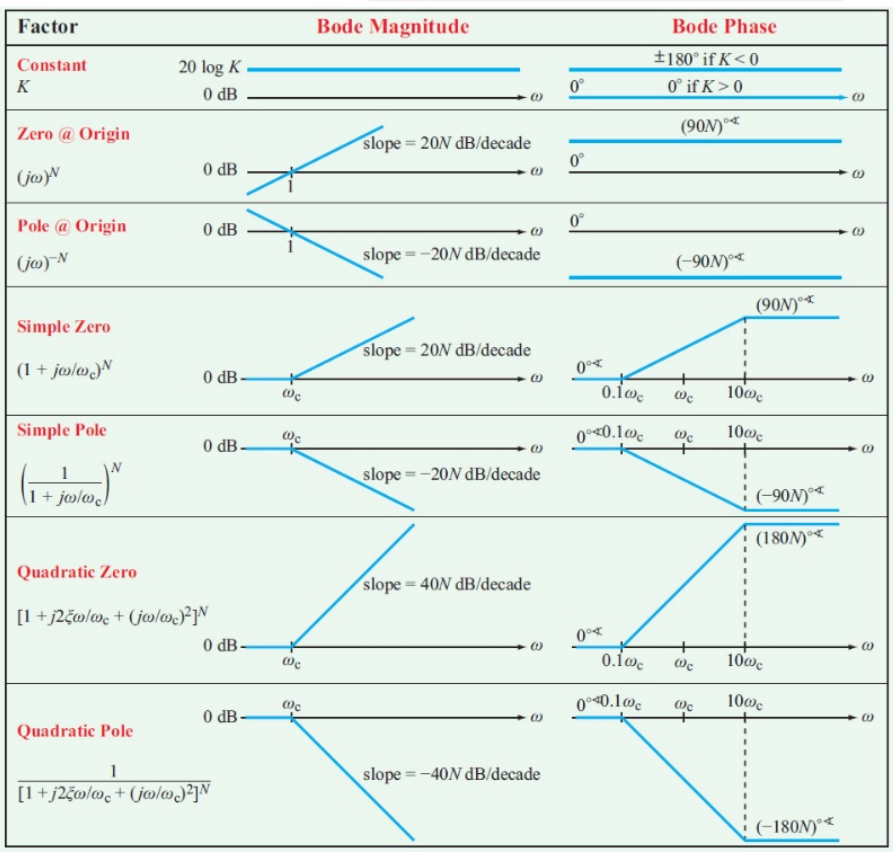
Grading
- 1 pt trying
- 2-3 pts partial
- 5 pts correct answer

Frequency-domain solution:

$Y(s) = \left(\frac{s}{s^2+1}\right)\left(\frac{s}{s^2+1}\right) = \frac{s^2}{(s^2+1)^2}$

NOT BIBO stable since the poles are on the j axis, so NOT in the open left-half plane.

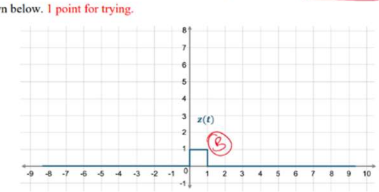




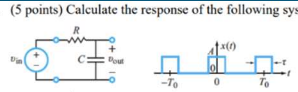
c. (5 points) If the signal $h_2(t)$ graphed above satisfies the equation: $h_2(t) = z(t) * h_1(t)$, where $*$ represents convolution, then graph the signal $z(t)$. Your final answer must be presented in the space below, and this will be the only answer graded. Ⓐ

Convolution of any signal $p(t)$ with $A\delta(t - T_0)$ results in $Ap(t - T_0)$, i.e. a scaled copy at $t = T_0$. Therefore, $z(t)$ is as shown below. 1 point for trying.

2 points for A and B



b. (5 points) Calculate the response of the following system to the input $x(t)$.



$x(t) = \frac{A\tau}{T_0} + \sum_{n=1}^{\infty} \frac{2A}{n\pi} \sin\left(\frac{n\pi\tau}{T_0}\right)\cos\left(\frac{2n\pi t}{T_0}\right)$

$H(\omega) = \frac{1/j\omega C}{R + 1/j\omega C} = \frac{1}{1+j\omega RC}$ Ⓐ $|H(\omega)| = \frac{1}{\sqrt{1+(\omega RC)^2}}$, $\angle H(\omega) = \tan^{-1}(\omega RC)$

$\Rightarrow y(t) = \frac{A\tau}{T_0}|H(0)| + \sum_{n=1}^{\infty} \frac{2A}{n\pi}|H(n\omega_0)| \sin\left(\frac{n\pi\tau}{T_0}\right)\cos\left(\frac{2n\pi t}{T_0} - \angle H(n\omega_0)\right)$

$= \underbrace{\frac{A\tau}{T_0}}_{Ⓑ} + \sum_{n=1}^{\infty} \underbrace{\frac{2A}{n\pi\sqrt{1+(n\omega_0 RC)^2}}}_{Ⓒ} \sin\left(\frac{n\pi\tau}{T_0}\right)\cos\left(\frac{2n\pi t}{T_0} - \underbrace{\tan^{-1}(n\omega_0 RC)}_{Ⓓ}\right)$

where $\omega_0 = \frac{2\pi}{T_0}$

Grading:
- 1 pt trying
- +1 pt each Ⓐ – Ⓓ